



AI Meal Planner and Smart Grocery Assistant

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Problem

Meal planning is time-consuming

Hard to balance:

- health goals
- budget
- dietary restrictions

Users must manually search recipes and plan meals



Solution

AI-powered meal planning web app

Users input:

- goal
- calories
- budget
- diet
- exclusions

App generates a full meal plan + grocery list

AI Meal Planner & Smart Grocery Assistant

Generate a personalized meal plan using real recipe data, nutrition information, budget goals, and AI-powered recommendations.

Meal goal

Healthy

Diet restriction

Vegetarian

Ingredients to avoid

Example: peanuts, seafood, mushrooms

Target calories per day

1000

Grocery budget (\$)

50

How many days?



Technologies

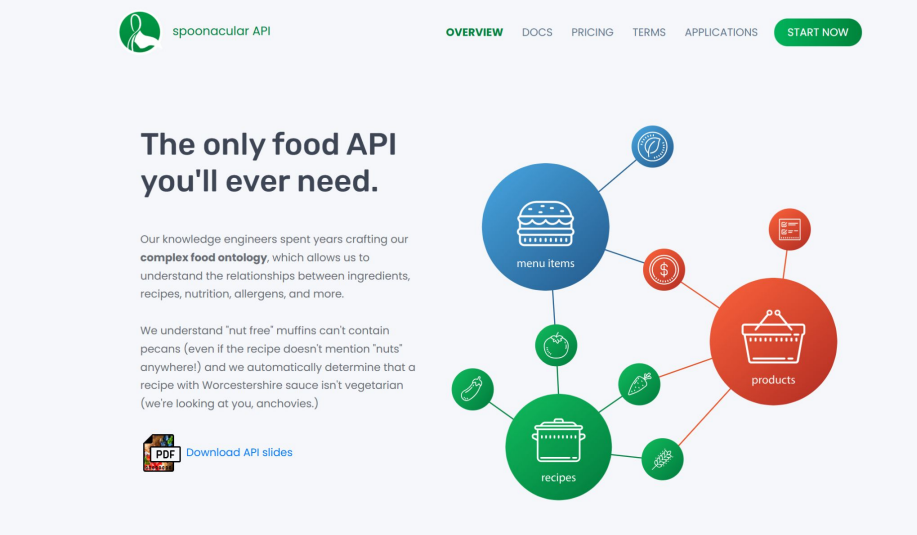
Python

Streamlit

Spoonacular API

OpenAI API

Combines real data + generative AI



The screenshot shows the Spoonacular API website. At the top, there is a navigation bar with the Spoonacular logo, the text "spoonacular API", and links for "OVERVIEW", "DOCS", "PRICING", "TERMS", "APPLICATIONS", and a "START NOW" button. The main heading reads "The only food API you'll ever need." Below this, a paragraph states: "Our knowledge engineers spent years crafting our **complex food ontology**, which allows us to understand the relationships between ingredients, recipes, nutrition, allergens, and more." A second paragraph provides an example: "We understand 'nut free' muffins can't contain pecans (even if the recipe doesn't mention 'nuts' anywhere!) and we automatically determine that a recipe with Worcestershire sauce isn't vegetarian (we're looking at you, anchovies.)" A "Download API slides" link with a PDF icon is present. On the right, a diagram illustrates the "complex food ontology" with interconnected nodes: "menu items" (blue circle with a burger icon), "products" (red circle with a shopping cart icon), and "recipes" (green circle with a pot icon). Smaller nodes include a coin, a dollar sign, a tomato, a carrot, a chili pepper, and a leaf, all connected by lines to show their relationships within the ontology.



How it Works

1. User enters preferences
2. App calls Spoonacular API
3. Retrieves recipes + ingredients
4. Sends data to OpenAI
5. AI generates:
 - summary
 - grocery list
 - scores
 - suggestions

AI Meal Scores

- **Nutrition Score: 9/10**
The meals are rich in vegetables, whole grains, and proteins, making them nutrient-dense.
- **Budget Score: 6/10**
While the meals are healthy, they go over budget and some ingredients may not be cost-effective.
- **Convenience Score: 7/10**
The meal prep times are reasonable, but baking muffins may require more effort compared to other meal prep methods.
- **Overall Score: 7/10**
The plan has strong nutrition and reasonable convenience, but the budget issue pulls down its overall score.

Grocery List

Produce:

- 1 1/2 cups diced carrots
- 3 cups diced assorted peppers (sweet red Italian, green and red bell)
- 1 zucchini, diced
- 2 yellow squash, diced
- 10 cloves of garlic, chopped
- 4 tomatoes, halved, seeded and cut into 1/2 inch dice (3 cups)
- 4 tomatoes, skins removed and diced
- 24 medium fresh basil leaves
- 1/2 cup celery (4 stalks)
- 1/2 cup green beans



Features

Multi-day meal planning

Diet restrictions

Ingredient exclusions

Budget tracking

AI scoring system

Download grocery list

Monday

Chive and dill muffins

Ready in 45 minutes

Servings: 10

Ingredients:

- 1 cup all purpose flour
- 1 cup yellow cornmeal
- 1 Tbs sugar
- 2 tsp baking powder
- 1 tsp salt
- ½ tsp baking soda
- ½ tsp cayenne pepper
- ¼ cup fresh chives and dill, chopped
- 1 ½ cups plain yogurt
- 2 large eggs
- ¾ Tbs butter, melted



Demo



Challenges

API limitations

Messy or missing data

Debugging errors

Environment setup issues



What Worked/Didn't

Worked:

- API + AI integration
- real-time meal generation

Didn't:

- some recipes missing data



Conclusion

Solves a real-world problem

Uses generative AI effectively

Fully functional MVP